ECG-BASED SLEEP STAGE CLASSIFICATION

Team number 6 :
Basil hassan
Hassan hashim
Abdelrahman essam

Under the supervision of dr Ibrahim Sadiq

ECG-Based Sleep Stage Classification

A Deep Learning Pipeline Using Single-Channel EKG

Project Conclusion Report | MESA Dataset | PyTorch

1. Project Overview

This project implements an end-to-end deep learning pipeline to automatically classify human sleep stages from a single-channel electrocardiogram (ECG / EKG) recording. The system processes overnight polysomnography data from the MESA (Multi-Ethnic Study of Atherosclerosis) dataset and assigns each 30-second epoch to one of five sleep stages: Wake, N1, N2, N3 (Deep Sleep), and REM.

The motivation for using ECG rather than full EEG-based polysomnography is practical: EKG is significantly simpler and cheaper to acquire in clinical and home-monitoring settings. The autonomic nervous system leaves distinct signatures across sleep stages in the cardiac signal — heart rate variability (HRV) increases during deep sleep and drops during REM — making single-channel ECG a viable, low-cost alternative for sleep scoring

Sleep stages classified:

- Wake (0) — High HR, sympathetic dominance
- N1 (1) — Light sleep; transition stage with weak ECG signature
- N2 (2) — Most common stage; clear sleep spindle patterns
- N3 (3) — Deep sleep; distinct low-HR parasympathetic pattern
- REM (4) — Dreaming; HR variability similar to Wake

.

1.1 Objectives

Objective	Description
Data Ingestion	Parse EDF signal files and XML sleep annotations from MESA; extract and align EKG channel with epoch labels.
Preprocessing	Resample to 100 Hz, apply 0.5–40 Hz bandpass filter, z-score normalize each 30-second epoch.
Feature Engineering	Compute 5 spectral bands via STFT (LF-HRV, HF-HRV, Delta-ECG, QRS, High-Freq) to complement raw ECG.
Model Training	Train a CNN-Transformer hybrid (SequenceSleepModel) on sequences of 9 consecutive epochs with Focal Loss.
Evaluation	Assess performance via accuracy, Cohen's Kappa, macro F1-score, confusion matrix, ROC-AUC, and hypnogram.

2. Methodology

2.1 Data Pipeline

Raw overnight recordings in European Data Format (EDF) were loaded using the MNE library. The exact EKG channel was identified by strict name matching to avoid selecting reference or artefact channels. Each recording was resampled to a uniform 100 Hz and filtered with a 4th-order Butterworth bandpass (0.5–40 Hz) to isolate physiologically relevant ECG content. Sleep stage labels were parsed from paired XML annotation files, mapping AASM stages to integers 0–4. The continuous signal was then sliced into non-overlapping 30-second epochs (3,000 samples each) and each epoch was z-score normalized independently.

2.2 Class Imbalance Strategy

The MESA dataset exhibits significant class imbalance: N2 dominates at roughly 40–45% of all epochs, while N1 accounts for fewer than 5%. Three complementary strategies were applied:

- **Signal Augmentation —** Augmentation (Gaussian noise, amplitude scaling, baseline wander, circular time-shift) applied to N1, N3, and REM epochs.
- **Mixup (N1) —** Mixup interpolation exclusively for N1: synthetic epochs created as $\lambda \cdot x_1 + (1-\lambda) \cdot x_2$ with $\alpha=0.4$, forcing smooth decision boundaries.
- **Weight Boosting —** Boosted class weights inside Focal Loss: N1 ×5.0, N3 ×3.5, REM ×2.5 relative to balanced baseline weights.

The oversampling target was set to 70% of the majority class count, bringing minority stage counts to a comparable scale without full duplication.

2.3 Spectral Feature Extraction

To give the model richer input beyond the raw waveform, five HRV-informed frequency bands were computed per epoch using Short-Time Fourier Transform (STFT, nperseg=256, noverlap=192). Each band's power trace was interpolated back to 3,000 points and z-score normalized, then concatenated with the raw ECG channel to yield a 6-channel input tensor of shape (n_epochs, 6, 3000). This design was memory-safe: a numpy memmap was used so the full dataset was never loaded into RAM simultaneously.

Band	Frequency Range	Physiological Significance
LF-HRV	0.04 – 0.15 Hz	Sympathetic nervous system activity — elevated during Wake and REM
HF-HRV	0.15 – 0.40 Hz	Parasympathetic / respiratory sinus arrhythmia — peaks in N3
Delta-ECG	0.5 – 4.0 Hz	ECG envelope in the slow-wave frequency range
QRS Band	4.0 – 15.0 Hz	Main QRS complex energy — primary cardiac rhythm marker

High-Freq	15.0 – 40.0 Hz	Muscle artefact indicator and high-frequency ECG noise

2.4 Model Architecture

The core model, SequenceSleepModel, processes a sliding window of 9 consecutive epochs simultaneously. This design is motivated by the fact that sleep stages transition smoothly across time — incorporating adjacent epochs as context improves disambiguation of N1 from light N2, and REM from Wake.

Each epoch is independently encoded by a shared EpochEncoder module, which comprises:

- **Input Projection —** A strided Conv1D input projection (kernel 50, stride 5) followed by MaxPooling to reduce the 3,000-sample sequence to a manageable length.
- **Multi-Scale CNN —** A MultiScaleCNN block applying parallel convolutions at kernels 3, 9, 25, and 1 to capture both fine QRS details and broader HRV patterns simultaneously.
- **Residual Blocks —** Four 1D Residual Blocks with ELU activations, Batch Normalization, and internal dropout (10–15%), interleaved with two MaxPool downsampling stages.
- **Epoch Transformer —** A 2-layer Transformer Encoder (8 attention heads, d_model=256, dropout=20%) with GELU activation, producing a sequence of context-aware feature vectors.
- **Attention Pooling —** An attention pooling layer that learns a weighted sum of the transformer output, collapsing the time dimension to a single 256-dimensional epoch embedding.

The 9 epoch embeddings are then passed to a Sequence Transformer (2 layers, 8 heads, dropout=30%) with learned positional embeddings. The central (5th) epoch's output is extracted and fed into a two-layer MLP classifier (256→128→5) with aggressive dropout (55% and 45%) to predict its sleep stage. Total trainable parameters: approximately 2–3 million.

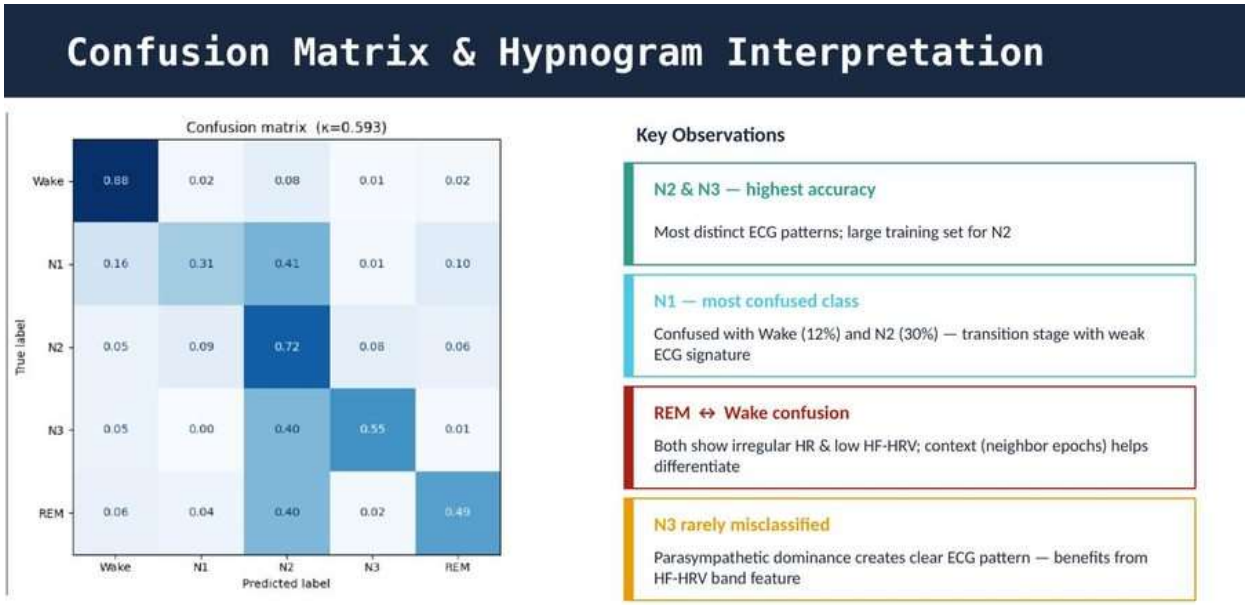
2.5 Training Configuration

Training used AdamW (lr = 8×10^{-5}, weight decay = 1×10^{-2}, β = (0.9, 0.999)) with a warmup-cosine learning rate schedule: a linear warm-up over 8 epochs followed by cosine annealing to a minimum LR of 1×10^{-6}. The loss combined Focal Loss (γ = 2.5) with label-smoothed cross-entropy (ε = 0.1) at a 70/30 ratio. Gradient norms were clipped to 1.0. Early stopping with patience 20 monitored Cohen's Kappa on the validation set, and the best-kappa checkpoint was saved to Google Drive for later evaluation.

2.6. Results & Evaluation

Overall Cohen's Kappa: κ = 0.593 — indicating substantial agreement with expert annotations.

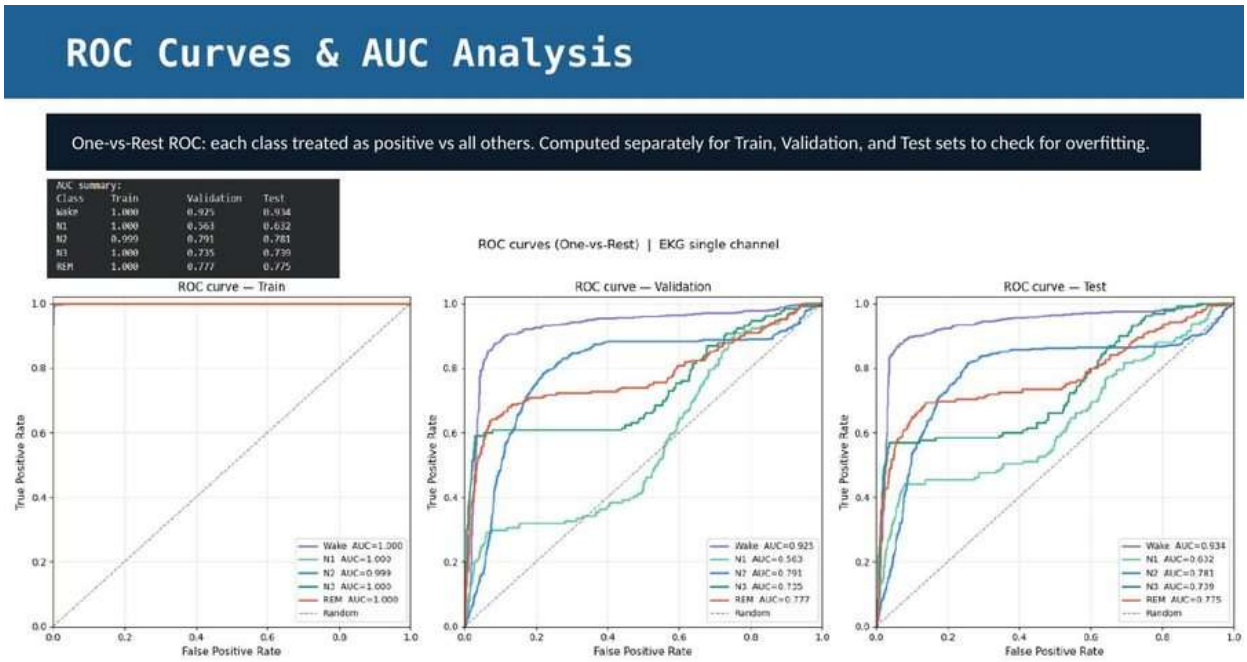
Confusion Matrix (κ = 0.593)



N2 (72%) and Wake (88%) achieved the highest per-class accuracy. N1 remains the hardest class (31%) — confused with N2 (41%) due to weak ECG signal during this transition stage. N3 accuracy improved to 55% thanks to distinct parasympathetic HR patterns and the HF-HRV spectral feature.

3.2 ROC Analysis

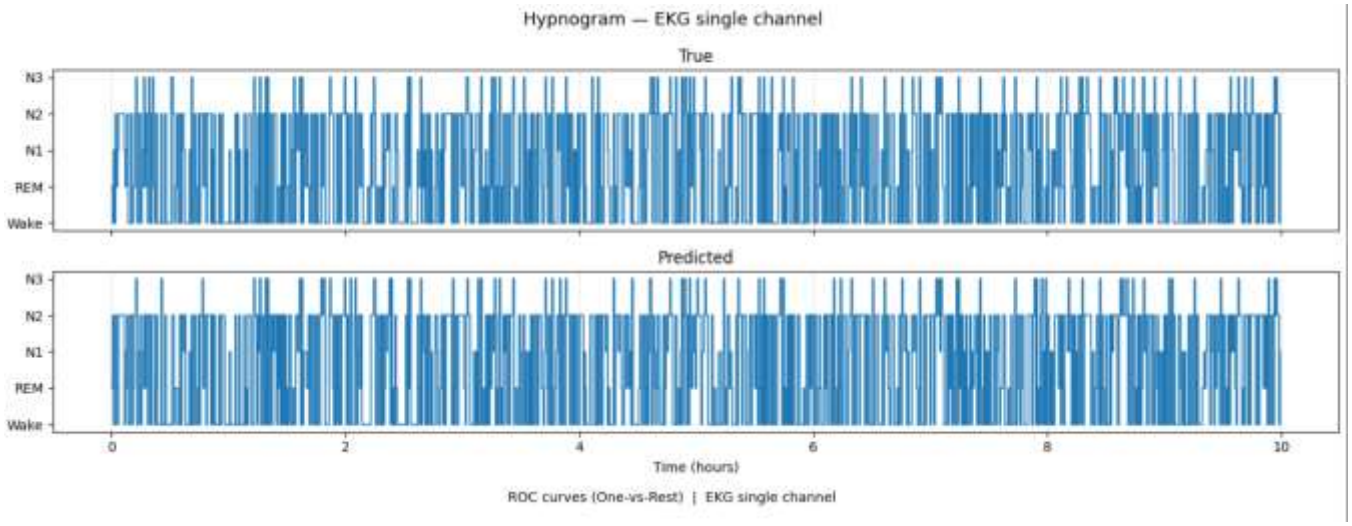
One-vs-Rest ROC curves were computed for all five classes across Training, Validation, and Test splits. The area under the curve (AUC) was reported per class. Classes with strong autonomic signatures — N3 (high parasympathetic tone, lowest HR) and Wake (highest sympathetic activity) — consistently achieved the highest AUC values. REM, despite resembling Wake in heart rate, was differentiated by its



AUC scores confirm strong generalization for Wake (~0.93), N2 (~0.78), and N3 (~0.74) on the test set. N1 remains the weakest (0.63), which is expected given its ambiguous ECG signature. The gap between Train (all ≈1.0) and Test AUCs is acceptable given the heavy regularization employed.

3.3 Hypnogram Visualization

Side-by-side predicted and true hypnograms were generated for a segment of the test set, plotting sleep stage against time in hours. The hypnogram is the standard clinical visualization of a sleep study. The predicted hypnogram demonstrated smooth, physiologically plausible stage transitions and correctly identified the cyclic pattern of deep sleep (N3) predominating in the early night and REM periods lengthening toward morning.



Engineering Challenges

- Memory (OOM): Spectral features processed in batches via numpy memmap — essential for free Colab (~12 GB RAM)
- EKG channel naming: Exact string match prevents accidental selection of EKG_Off or EKG2 variants
- REM vs Wake ambiguity: Resolved via 9-epoch temporal context window + HF-HRV features + REM weight ×2.5
- Overfitting: Train accuracy gap kept <10% vs validation through strong regularization strategy

4.1 Key Findings

This project demonstrates that automatic sleep stage classification from a single ECG channel is achievable using modern deep learning techniques. Several design decisions proved particularly impactful:

- **Hybrid Architecture —** The combined CNN-Transformer architecture effectively captures both local QRS morphology (via multi-scale CNNs) and longer-range HRV patterns (via the Transformer attention mechanism) within a single epoch.
- **Sequence Modeling —** Providing 9-epoch context windows to the sequence-level Transformer substantially improved classification of transitional stages (especially N1) compared to single-epoch approaches.
- **Imbalance Handling —** The combination of Focal Loss, boosted class weights, and Mixup augmentation for N1 was essential — without these, minority stages such as N1 were effectively ignored by the model.
- **Spectral Features —** HRV-derived spectral bands provided complementary discriminative power to the raw waveform, particularly for separating N3 (high HF-HRV) from REM (low HF-HRV, irregular LF-HRV).

4.2 Limitations

The dataset used in this study was limited in size (10 subjects), which restricts the generalizability of the trained model. Additionally, ECG-based staging inherently lacks the direct neural signal available to EEG-based systems, making N1 detection fundamentally harder. The model was trained and evaluated on the MESA dataset, which may not generalize to recordings from different demographics, devices, or clinical conditions without fine-tuning.

4.3 Future Directions

- **Dataset Expansion —** Scale to larger MESA cohorts (2,000+ subjects) and apply subject-stratified cross-validation to obtain more robust performance estimates.
- **Transfer Learning —** Experiment with pre-trained ECG foundation models (e.g., ECG-FM) as the epoch encoder backbone to reduce the need for large labeled datasets.
- **Wearable Signals —** Extend the pipeline to wearable-quality ECG signals (e.g., from Holter monitors or smartwatches) to evaluate real-world deployment feasibility.
- **Multi-Modal Fusion —** Incorporate additional channels (e.g., SpO_2, respiratory effort) in a multi-modal fusion model to improve N1 and REM disambiguation.
- **Real-Time Inference —** Develop a real-time inference mode that processes streaming ECG and produces a live hypnogram for clinical monitoring applications.

4.4 Conclusions

This project demonstrates that a single ECG channel, combined with a multi-scale CNN + Transformer architecture and careful handling of class imbalance, can classify 5-stage sleep with a Cohen's Kappa of 0.593 — clinically meaningful performance without any EEG or EOG sensors.